

论文阅读：RadioSim Agent-让大模型自己调用射线追踪分析信号覆盖

RadioSim Agent 相当于 把昨天的MAPS的“AI+ 仿真器”思路从电路分析搬到了通信领域

让大模型自己调用射线追踪来分析无线电信号覆盖。

一.概述：

RadioSim Agent = 让AI听懂人话 → 自己调用射线追踪工具算信号 → 生成覆盖图 + 告诉你哪强哪弱

propagation simulations and visual analyses through callable tool interfaces.

A. Framework Overview

As shown in Fig. 1, the framework comprises four main components:

(1) **Natural Language Planner:** The planner serves as the reasoning core of the RadioSim Agent and is implemented using the OpenAI GPT-4o-mini model. It interprets user instructions expressed in natural language and translates them into structured simulation or analysis tasks. Specifically,

The underlying ray-tracing engine computes direct & reflected propagation paths between the transmitter and grid of receiver points, producing deterministic pathloss & associated feature maps [8], [9]. For receiver locations where no valid ray path is available, the module applies a 3GPP-based empirical pathloss model [10] to approximate non-line-of-sight (NLOS) attenuation, ensuring full spatial coverage. In addition, building entry loss (BEL) is computed following ITU-R P.2109 recommendations [11] to account for signal penetration through walls.

All these computations are encapsulated within a Python-based callable tool, `simulate_radio_environment`.

就像MAPS让AI看懂电路图、调用SPICE算电路一样，

只不过这次换成了通信领域的“射线追踪”

二.背景：为什么要做这件事？

1. 传统电磁仿真的痛点

痛点	具体表现
门槛高	射线追踪仿真需要专家配置参数，新手不会用
流程死板	点按钮、填表单、跑程序，步骤固定且繁琐
结果难懂	给你一张花花绿绿的热力图，不告诉你哪里信号好、为什么好

痛点	具体表现
缺乏交互	想换个场景？重新来一遍，不能"对话式"调整

Accurate radio propagation modeling is fundamental to the design, optimization, and analysis of emerging 6G wireless communication systems [1]. Deterministic electromagnetic (EM) simulators, such as ray-tracing-based solvers, can provide accurate predictions of signal propagation. However, they often demand expert configuration, follow rigid workflows, and offer limited semantic interpretability and automation.

2.现有 Agent 研究的局限

之前的 Agent 工作（如 WirelessAgent、A-Core）主要做**网络管理层**的事（资源分配、网络切片），**没有真正去做物理层的电磁仿真计算。**

Despite these advances, existing agentic and LLM-driven frameworks primarily operate at the network-management or data-analysis level. They optimize configurations or interpret measurements but do not *simulate, analyze, or explain* radio propagation using first-principles EM computation. As a result,

概念解释：

物理层 = 管"信号怎么在空气中跑"（修路+开车）

网络管理层 = 管"网络怎么高效运转"（调度+管理）

RadioSim Agent 的亮点 = 之前Agent只会调度，现在Agent还会修路

三.RadioSim Agent 的做法具体过程

Step 1: LLM 规划器理解需求 → 提取参数

你的输入	LLM 提取的参数
"模拟慕尼黑01场景，无人机在(100,100,15)"	环境= munich01 , 坐标=(100,100,15)
"50×50的接收网格"	网格大小=50×50

你的输入	LLM 提取的参数
"考虑所有传播机制"	LOS=True, REF=True, GREF=True, NLOS=True, BEL=True

them into structured simulation or analysis tasks. Specifically, the planner extracts relevant parameters such as environment name, transmitter coordinates, receiver grid size, and selected propagation mechanisms. Leveraging its reasoning capability,

如果指令模糊，LLM 会主动提问（比如“请指定网格分辨率”），确保理解正确。

Step 2：仿真工具库执行计算 → 生成路径损耗图

核心函数： `simulate_radio_environment()`

一个 Python 可调用的工具函数，接受参数、运行射线追踪、输出路径损耗矩阵。

(2) Simulation Tool Library: The Simulation Tool Library provides deterministic EM modeling functions for computing key propagation mechanisms, including line-of-sight (LOS), wall reflections (REF), and ground reflections (GREF). The

传播机制	含义	计算方式
LOS	直射	射线追踪，看基站和接收点之间有没有遮挡
REF	墙面反射	追踪射线碰到建筑物墙面后的反射路径
GREF	地面反射	考虑地面反射对信号的影响
NLOS	非视距	没有有效射线时，用 3GPP 经验模型估算
BEL	建筑穿透损耗	信号穿过建筑物墙壁的衰减（ITU-R P.2109 标准）

Step 3：视觉推理工具分析热力图 → 生成语义描述

RadioSim 的仿真结果是图片（热力图），需要“看懂”：

els. In the *Observe* phase, the agent analyzes and summarizes the generated outputs. When visual interpretation is needed, the Vision Reasoning Tool is invoked to extract insights directly from the pathloss heatmaps, providing a multimodal summary that combines visual and textual reasoning. This uni-

视觉分析内容	示例输出
信号强度范围	"路径损耗从 110dB（强信号）到 170dB（弱信号）"
空间分布	"强信号集中在左下角，弱信号在右上角"
异常区域	"墙面和反射表面附近有明显梯度变化"
遮挡识别	"建筑物后方信号衰减显著"

Step 4：整合输出 → 最终回答

把仿真参数、数值结果、视觉分析整合成一段自然语言报告

四.关键实验结果

实验：用一句话指令，它能自己跑完整个流程吗？

We evaluated the agent’s ability to autonomously run radio propagation simulations using the following natural language prompt:

“Simulate pathloss in the Munich01 scenario with a UAV at (100, 100, 15) over a 50×50 receiver grid considering all propagation mechanisms, and provide a concise technical summary of the resulting pathloss heatmap.”

AI的执行过程

步骤	AI在做什么
Thought	“用户要求在慕尼黑场景仿真，需要调用仿真工具”
Action	调用 <code>simulate_radio_environment()</code>
Observation	“仿真完成，信号图保存在 data/pathloss.png”
Action	调用 <code>summarize_pathloss_image()</code> 看图分析

步骤	AI在做什么
Observation	“信号范围110-170dB，左下角强，右上角弱，墙边衰减明显”
Final Answer	输出最终总结

结果：成功了。AI自己完成了从“听懂指令”到“输出分析”的全流程。